# DEEP Metrics

## Eastern Corn Belt Plains reference assessment (eastern-corn-belt-plains@6)

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|---|---|
| Stream | Mink Brook |
| Coordinates | 43.68581234, -72.23671234 |
| COMID / HUC8 | 9327042 / 01080104 |
| Drainage area | 12.43 km2 |
| Reach length | 1000.0 ft |
| Watershed basis | HR reach watershed (STAF site engine v0.5.0) |
| StreamCat reach | COMID 9327042 (this reach) |

| Function | Metric | F/D | How it is measured | Status | Value | Source |
|---|---|---|---|---|---|---|
| Catchment hydrology | Agriculture, crop and hay (%) | D | Percent of the delineated watershed in cultivated crops plus hay and pasture, computed by DEEP. No field measurement. | Available | 3.0 | NLCD 2021 |
| Catchment hydrology | Impervious surface (%) | D | Percent of the delineated watershed in impervious surface, computed by DEEP. No field measurement. | Available | 12.75 | EPA StreamCat |
| Surface water storage | Wetland (%) | D | Percent of the watershed in woody plus herbaceous wetland (NLCD 2019), read from EPA StreamCat for the reach. No field measurement. | Reference only | 0.4 | Computed by DEEP |
| Reach inflow | Road density (km/sq km) | D | Road length per square kilometer of the delineated watershed, computed by DEEP. No field measurement. | Available | 3.0 | NLCD 2021 |
| Streamflow regime | Base flow index (%) | D | Percent of streamflow that comes from base flow in the watershed (USGS base flow index grid), read from EPA StreamCat for the reach. No field measurement. | Available | 12.75 | EPA StreamCat |
| Streamflow regime | Degree of regulation (%) | D | Normal storage of upstream dams as a percent of mean annual runoff, both read from EPA StreamCat for the reach. No field measurement. | Available | 0.4 | Computed by DEEP |
| Low flow and baseflow dynamics | Bankfull width:depth ratio | F | At each transect, measure bankfull width, bankfull height above the water surface and thalweg depth. Value: mean bankfull width divided by mean bankfull depth (bankfull height plus thalweg depth). | Entered | 7.5 | |
| Low flow and baseflow dynamics | Residual pool depth (cm) | F | Measure thalweg depth at 10 or 15 evenly spaced stations between each pair of transects (100 or 150 per reach). Value: mean residual pool depth in cm, from that profile and the reach slope (NRSA physical habitat calculations). | *Measure in the field* | | |
| Low flow and baseflow dynamics | Base flow index (%) | D | Percent of streamflow that comes from base flow in the watershed (USGS base flow index grid), read from EPA StreamCat for the reach. No field measurement. | Available | 12.75 | EPA StreamCat |
| High flow dynamics | Impervious surface (%) | D | Percent of the delineated watershed in impervious surface, computed by DEEP. No field measurement. | Available | 12.75 | EPA StreamCat |
| Floodplain connectivity | Bankfull width:depth ratio | F | At each transect, measure bankfull width, bankfull height above the water surface and thalweg depth. Value: mean bankfull width divided by mean bankfull depth (bankfull height plus thalweg depth). | Entered | 7.5 | |

| Function | Metric | F/D | How it is measured | Status | Value | Source |
|---|---|---|---|---|---|---|
| Floodplain connectivity | Bankfull height above channel (m) | F | At each transect, measure the bankfull height above the present water surface in m, with a rod at the water's edge and a clinometer held level at bankfull. Value: mean of the 11 heights. | *Measure in the field* | | |
| Hyporheic connectivity | Sand + fines (%) | F | Record the particle size class at 5 evenly spaced points on each of the 11 transects and the 10 cross-sections midway between them (105 particles). Value: percent sand or finer. | *Not applicable* | | |
| Hyporheic connectivity | Embeddedness (%) | F | At 5 evenly spaced points on each of the 11 transects, estimate the percent of the particle buried in sand or finer sediment. Value: mean of the 55 estimates. | *Measure in the field* | | |
| Channel evolution | Bankfull width:depth ratio | F | At each transect, measure bankfull width, bankfull height above the water surface and thalweg depth. Value: mean bankfull width divided by mean bankfull depth (bankfull height plus thalweg depth). | Entered | 7.5 | |
| Channel evolution | Bankfull height above channel (m) | F | At each transect, measure the bankfull height above the present water surface in m, with a rod at the water's edge and a clinometer held level at bankfull. Value: mean of the 11 heights. | *Measure in the field* | | |
| Channel and floodplain dynamics | Sinuosity | F | Take a compass backsight bearing and the distance between each pair of neighboring transects. Value: channel length along those segments divided by the straight-line distance between the reach ends. | Entered | 12.5 | |
| Channel and floodplain dynamics | Bank angle (degrees) | F | On both banks of each transect, lay a pole about 1 m long against the bank and read its angle from horizontal with a clinometer, in degrees. Value: mean of the 22 readings. | *Measure in the field* | | |
| Sediment continuity | Relative bed stability (log) | F | Computed: the base 10 log of the geometric mean particle diameter divided by the largest diameter the bankfull flow can move, from the particle count, reach slope, bankfull depth, wood tally and residual pools (NRSA physical habitat calculations). | *Measure in the field* | | |
| Sediment continuity | Sand + fines (%) | F | Record the particle size class at 5 evenly spaced points on each of the 11 transects and the 10 cross-sections midway between them (105 particles). Value: percent sand or finer. | *Not applicable* | | |
| Bed composition and bedform dynamics | Substrate size (log mm) | F | From the 105-particle size class count (as for sand and fines). Value: base 10 log of the geometric mean particle diameter in mm, from the size classes (NRSA physical habitat calculations). | *Measure in the field* | | |
| Bed composition and bedform dynamics | Large wood volume (m3/100m) | F | Between each pair of transects, tally every piece of wood at least 10 cm in diameter and 1.5 m long in or above the bankfull channel, by length and diameter class. Value: wood volume per 100 m of channel, from the tally. | *Measure in the field* | | |
| Bed composition and bedform dynamics | Embeddedness (%) | F | At 5 evenly spaced points on each of the 11 transects, estimate the percent of the particle buried in sand or finer sediment. Value: mean of the 55 estimates. | *Measure in the field* | | |
| Light & thermal regime | Canopy density, mid-channel (%) | F | At mid-channel on each transect, read a convex spherical densiometer facing upstream, downstream, left and right (covered points of 17). Value: mean of the 44 readings, as a percent. | Entered | 17.5 | |

| Function | Metric | F/D | How it is measured | Status | Value | Source |
|---|---|---|---|---|---|---|
| Carbon processing | Riparian veg cover, woody (frac) | F | In a 10 m by 10 m plot on each bank at each transect, estimate the areal cover class of woody vegetation in the canopy, understory and ground layer. Value: summed woody cover of the 3 layers, averaged over the 22 plots, as a fraction. | *Measure in the field* | | |
| Habitat provision | Fast-water / riffle habitat (%) | F | Percent fast-water (riffle/run) habitat; more generally better for many biota. Ecoregion-dependent. | *Measure in the field* | | |
| Habitat provision | Natural fish cover (frac) | F | In the 10 m of channel centered on each transect, estimate the areal cover class of each fish cover type on the form (wood, brush, overhanging vegetation, undercut banks, boulders, others). Value: summed cover of the natural types, averaged over the transects. | *Measure in the field* | | |
| Watershed connectivity | Mapped dams within one mile (dams) | D | Dams in the USACE National Inventory of Dams within 1 mile of the site, counted by DEEP. Note in the field any barrier the inventory does not show. | Available | 0.4 | Computed by DEEP |
| Community dynamics | EPT taxa richness | | | *Not scored* | | Insufficient reference support |
| Community dynamics | Shannon diversity (H') | | | *Not scored* | | Insufficient reference support |
| Community dynamics | Tolerant individuals (%) ~ HBI | | | *Not scored* | | Insufficient reference support |
| Population support | Benthic total taxa richness | | | *Not scored* | | Insufficient reference support |
| Light and thermal regime | Chlorophyll a (ug/L) | | | *Not scored* | | Insufficient reference support |
| Water and soil quality | Specific conductivity (uS/cm) | | | *Not scored* | | Insufficient reference support |
| Nutrient cycling | Dissolved nitrogen (mg N/L) | | | *Not scored* | | Insufficient reference support |
| Water and soil quality | pH (Std. Units) | | | *Not scored* | | Insufficient reference support |
| Nutrient cycling | Total phosphorus (ug/L) | | | *Not scored* | | Insufficient reference support |
| Water and soil quality | Turbidity (NTU) | | | *Not scored* | | Insufficient reference support |
| Population support | Native fish taxa richness | | | *Not scored* | | Insufficient reference support |

F is measured in the field and D is answered from the desk. Available: the value was pulled and is printed on the field worksheet. Unavailable: no source answered, so measure or estimate it on site. Reference only: the value is shown and kept out of the score, because the curve was fitted on another data source. Not scored: the assessment withholds the metric for insufficient reference support.